

HAICI YANG

haici.yang@gmail.com
<https://haiciyang.github.io/>

Sr. Researcher @ Dolby
Atlanta, Georgia

Audio Coding, Speech Generation, Music Source Separation, Spatial Audio

EDUCATION

PhD	Indiana University Bloomington, IN. Intelligent System Engineering Advised by Professor Minje Kim	2019 – Current
MS	Peking University Beijing, China Informatics Advised by Professor Jun Wang in KVision Digital Library Lab. Visiting student (2018-2019) at Indiana University, advised by Professor Katy Boner	2017 – 2019
BS	Beijing Normal University Beijing, China Information System; Minored in Psychology GPA 4.2/5; Graduated with Honor	2013 – 2017

WORK EXPERIENCE

Dolby Laboratories. June 2025 – Current
Sr. Researcher – Sound Lab. Atlanta, GA

Mitsubishi Electric Research Lab (MERL) July 2024 – Feb 2025
Research Intern, Cambridge, MA

- Supervisor: Gordon Wichern
- Generative speech enhancement [C-10] and Anomaly Sound Detection [C-9]

Indiana University Jan 2021 – Current
Research Assistant, Bloomington, IN

- Supervisor: Minje Kim
- Project 1: Source-aware neural codec targeting the mixture signal coding problem. Work published [C-2].
- Project 2: End-to-end music remixing model that jointly optimizes remixing and source separation tasks. Work published [C-3].
- Project 3: Low-latency predictive neural codec. Work published [C-4].
- Project 4: Generative neural audio codec. Work published [C-6, C-7]. Journal extension in progress.

Adobe Research Jun 2023 – Dec 2023
Research Intern. San Francisco, CA

- Supervisor: Jiaqi Su, Zeyu Jin
- Project: Generative speech enhancement on discrete speech tokens. Work published [C-8].

Microsoft – Microsoft Research Asia Nov 2022 – Mar 2023
Research Intern. Beijing, China

- Supervisor: Xiulian Peng
- Project: Speech coding.

Amazon.com, Inc – Lab126

Summer 2021

Applied Scientist Intern. Cambridge, MA

- Supervisor: Wontak Kim, Sanna Wager, Spencer Russell
- Project: Blind stereo-to-5-channel up-mixing involving spatial image and musical content disentanglement. Work published [C-4].

Amazon.com, Inc - Alex Speech Team

Summer 2020

Applied Scientist Intern. Sunnyvale, CA

- Supervisor: Oguz Elibol, Hongda Mao
- Project: An overview of the state-of-the-art speaker recognition model and exploring the application of Non-local Convolutional Neural Networks (NLCNN) in the speaker recognition problem.

Cyberinfrastructure Network Science Center

Aug 2018 – Mar 2019

Research Assistant. Bloomington, IN

- Supervisor: Katy Borner
- Project: Mapping the co-evolution of artificial intelligence, robotics, and the Internet of Things over 20 years through keyword detection, network analysis, and visualization. Work published [J-1]

WWF China

Summer 2018

Analyst Intern & Volunteer. Beijing China

- Website design and development for the Human Resource Team.

Nielsen

Summer 2016

Analyst Assistant. Beijing China

- Data-driven user study reports produced for the Telecom Team

PUBLICATIONS

Conference Papers

[C-10] **Haici Yang**, Gordon Wichern, Ryo Aihara, Yoshiki Masuyama, Sameer Khurana, François G. Germain, Jonathan Le Roux. "Investigating Continuous Autoregressive Generative Speech Enhancement." *In Proc. Interspeech*, 2025.

[C-9] Kevin Wilkinghoff, **Haici Yang**, Janek Ebberts, François G Germain, Gordon Wichern, Jonathan Le Roux. "Keeping the Balance: Anomaly Score Calculation for Domain Generalization." *In Proc. ICASSP*, 2025.

[C-8] **Haici Yang**, Jiaqi Su, Minje Kim, and Zeyu Jin. "Genhancer: High-Fidelity Speech Enhancement via Generative Modeling on Discrete Codec Tokens." *In Proc. Interspeech*, 2024.

[C-7] **Haici Yang**, Inseon Jang, and Minje Kim. "Generative De-Quantization for Neural Speech Codec via Latent Diffusion." *In Proc. ICASSP*, 2024.

[C-6] Inseon Jang, **Haici Yang**, and Minje Kim. "Personalized Neural Speech Codec." In *Proc. ICASSP*, 2024.

[C-5] **Haici Yang**, Wootae Lim and Minje Kim, "Neural Feature Predictor and Discriminative Residual Coding for Low-Bitrate Speech Coding," In *Proc. ICASSP*, 2023.

[C-4] **Haici Yang**, Sanna Wager, Spencer Russell, Mike Luo, Minje Kim, and Wontak Kim. "Upmixing via Style Transfer: A Variational Autoencoder for Disentangling Spatial Images and Musical Content." In *Proc. ICASSP*, 2022.

[C-3] **Haici Yang**, Shivani Firodiya, Nicholas J. Bryan, and Minje Kim. "Don't Separate, Learn To Remix: End-To-End Neural Remixing With Joint Optimization." In *Proc. ICASSP*, 2022.

[C-2] **Haici Yang**, Kai Zhen, Seungkwon Beack, and Minje Kim. "Source-Aware Neural Speech Coding for Noisy Speech Compression." In *Proc. ICASSP*, 2021.

[C-1] Sunwoo Kim, **Haici Yang**, and Minje Kim. "Boosted Locality Sensitive Hashing: Discriminative Binary Codes for Source Separation." In *Proc. ICASSP*, 2020.
'Nomination of Best Student Paper Award'

Journal Papers

[J-3] Pooneh Mousavi, et al. "Discrete Audio Tokens: More Than a Survey!" TMLR.
Equal Contribution, Under Peer Review

[J-2] Kevin Wilkinghoff, et al. "Local Density-Based Anomaly Score Normalization for Domain Generalization." IEEE Transactions on Audio, Speech and Language Processing.
Under Peer Review

[J-1] Katy Börner, et al. "Mapping the Co-Evolution of Artificial Intelligence, Robotics, and the Internet of Things Over 20 Years (1998-2017)." *PLoS ONE*, 2020.

Domestic Publications (In Chinese)

[JC-2] **Haici Yang**, Jun Wang. "Construction and Visualization of Knowledge Graph on Academic Relationships among People in Song Dynasty." *Data Analysis and Knowledge Discovery*, 2019. (Demo: kvlab.org/cbdb_kg)

[JC-1] Jian Jin, **Haici Yang**. "A Topic Relevance Aware Model for Reviewer Recommendation." *Digital Library Forum*, 2017.

Thesis

[T-2] **Haici Yang**. "Latent Generative Models in Ultra-Low-Bitrate Audio Coding." *Doctorate Thesis*, Indiana University. 2025
In progress

[T-1] **Haici Yang**. "Network Extraction and Analysis on <Song Yuan Xue An> - Word Embedding Application on Chinese Traditional Text." *Master Thesis*, Peking University. 2019

Patents

[P-4] Method and apparatus for encoding/decoding neural network-based personalized speech. US Patent App. 18/886,296.

[P-3] Method and device for encoding/decoding audio signal based on dequantization through potential diffusion. US Patent App. 18/886,765.

[P-2] Speech coding method and apparatus for performing the same. US Patent App. 18/474,997.

[P-1] Methods of encoding and decoding speech signal using neural network model recognizing sound sources, and encoding and decoding apparatuses for performing the same. US Patent App. 17/326,035

TEACHING EXPERIENCE

Indiana University, Bloomington, IN

Fall 2019

Associated Instructor

- ENGR-E533 Deep Learning System